

PY1010 — Physics for Biological and Chemical Sciences

Summer 2025 Examination · Paper 1 · Full Marking Scheme

Module	PY1010 — Physics for Biological and Chemical Sciences
Internal examiners	Dr M. Kennedy, Prof. A. Ruth
Institution	University College Cork
Session	Summer 2025, Paper 1
Duration	3 hours (+ 15 minutes reading time)
Structure	Six questions, each given as an either/or choice (A or B). Answer one option of each of the six questions and all parts of that option. Marks for each part are shown in square brackets.
Materials	Non-programmable calculator permitted; constants on the paper's "useful information" page

Coverage note. This scheme works through **both options (A and B) of all six questions** in full. On the day you only choose one option per question, but working both lets you pick the one you're strongest on.

Source basis. Every answer is built **only** from the PY1010 lecture material given to students: the lecture slides (mechanics, fluids, thermodynamics) and the handwritten lecture notes (gas laws, electricity, magnetism, waves and optics). No textbook or outside material is used, so every method here is one you were taught in the module. Where the paper asks for something the lecture notes don't state explicitly, a highlighted note says so at that part (Q4(B)(a) and Q6(A)(d)).

Own-figure rule. Where a part carries forward a number from an earlier part, an answer that uses the correct method on an earlier wrong value still earns the method marks for the later part. Markers (and you, self-marking) should apply this throughout.

Numerical conventions. $g = 9.81 \text{ m s}^{-2}$ and the other constants are taken from the paper's useful-information page. Intermediate values are kept unrounded, and final answers are given to 2–3 significant figures. Answers that differ only in the last digit through rounding (e.g. using $g = 9.8$, as some lecture examples do) should receive full marks.

Study-8

Question 1 · Option A

Speed vs velocity, area, density and mass flow of blood — 10 marks

Given: Blood: average speed 1.4 km/hr, density 1060 kg/m³; aorta radius 1.2 cm.

1(A)(a)

[1 mark]

Express the average speed of blood in m/s.

ANSWER

$$v = 0.39 \text{ m/s (0.389 m/s)}$$

WORKING

Convert using conversion factors (1 km = 1000 m, 1 hr = 3600 s):

$$v = 1.4 \text{ km/hr} \times (1000 \text{ m} / 1 \text{ km}) \times (1 \text{ hr} / 3600 \text{ s}) = 1400 \text{ m} / 3600 \text{ s}$$

EXAM TECHNIQUE

Keep the unrounded value (0.3889 m/s) in your calculator for parts (d) and (e). Rounding to 0.39 early changes the final answers slightly.

Marks

Awarded for

1

Correct conversion to 0.39 m/s **with unit**.

1(A)(b)**[2 marks]**

What is the difference between speed and velocity?

ANSWER

Speed is a scalar (distance/time); velocity is a vector (displacement/time) with magnitude and direction.

WORKING

Speed = total distance ÷ total time. It is a magnitude only (a scalar) and says nothing about direction.

Velocity = displacement ÷ time ($v = \Delta x / \Delta t$). Displacement is a vector, so velocity has both a **magnitude and a direction** and depends on the coordinate system chosen.

Example: a car driving a full lap back to its start has a non-zero average speed but zero average velocity, because its displacement is zero.

Marks	Awarded for
1	Speed defined as distance/time with no direction (scalar).
1	Velocity defined as displacement/time with magnitude and direction (vector).

1(A)(c)**[2 marks]**

Calculate the cross sectional area of a typical aorta, giving your answer in m².

ANSWER

$$A = 4.5 \times 10^{-4} \text{ m}^2 \text{ (} 4.52 \times 10^{-4} \text{ m}^2 \text{)}$$

WORKING

Circular cross-section, $A = \pi r^2$, with $r = 1.2 \text{ cm} = 0.012 \text{ m}$:

$$A = \pi \times (0.012 \text{ m})^2 = \pi \times 1.44 \times 10^{-4} \text{ m}^2$$

Marks	Awarded for
1	Correct conversion of r to metres and use of $A = \pi r^2$.
1	Correct answer in m ² .

1(A)(d)**[3 marks]**

Calculate the mass of blood which leaves an aorta in 1 second.

ANSWER **$m \approx 0.19 \text{ kg of blood per second (0.186 kg)}$** **WORKING**

In 1 s the blood moves a distance $x = v t = 0.3889 \text{ m}$. So it fills a cylinder of cross-section A and length x :

$$V = A \times x = 4.524 \times 10^{-4} \text{ m}^2 \times 0.3889 \text{ m} = 1.759 \times 10^{-4} \text{ m}^3$$

Then use density $\rho = m/V$, so $m = \rho V$:

$$m = 1060 \text{ kg/m}^3 \times 1.759 \times 10^{-4} \text{ m}^3 = 0.186 \text{ kg}$$

Marks	Awarded for
1	Distance travelled in 1 s ($x = vt$) or volume per second ($V = Av$).
1	Correct volume of blood.
1	Correct mass via $m = \rho V$, with unit.

1(A)(e)**[2 marks]**

45% of the blood is red blood cells; average mass of a red blood cell is 27×10^{-15} kg. Calculate the number of red blood cells which leave an aorta in 1 second.

ANSWER **$N \approx 3.1 \times 10^{12}$ red blood cells per second****WORKING**

Mass of red blood cells per second = 45% of the answer to (d):

$$m_{\text{RBC}} = 0.45 \times 0.1865 \text{ kg} = 0.0839 \text{ kg}$$

Number of cells = total RBC mass $\div$ mass of one cell:

$$N = 0.0839 \text{ kg} \div (27 \times 10^{-15} \text{ kg}) = 3.1 \times 10^{12}$$

EXAM TECHNIQUE

The question doesn't say whether the 45% is by mass or by volume. Taking it as 45% of the mass from (d) is the natural reading. If you take it by volume, state that you assume the red cells have the same density as whole blood, and the answer is the same.

Marks Awarded for

1 45% applied to the mass from (d).

1 Correct division, giving $\approx 3.1 \times 10^{12}$ (own-figure from (d) accepted).

Question 1(A) Total: 10 Marks

Question 1 · Option B

Acceleration and force, glacier kinematics, friction — 10 marks

1(B)(a)

[2 marks]

Define and give the SI units for (i) acceleration, and (ii) force.

ANSWER

$$\mathbf{a = \Delta v / \Delta t \text{ [m s}^{-2}\text{]}; F = ma \text{ [N = kg m s}^{-2}\text{]}}$$

WORKING

(i) **Acceleration** is the change in velocity divided by the time required for the change, $a = \Delta v / \Delta t$. It is a vector. SI unit: **m s⁻²**.

(ii) **Force**: by Newton's second law, the net force equals the rate of change of momentum, $F = dp/dt = ma$. A net force causes a mass to accelerate. SI unit: the **newton**, $1 \text{ N} = 1 \text{ kg m s}^{-2}$. This is the force that gives 1 kg an acceleration of 1 m s^{-2} .

Marks	Awarded for
1	Acceleration defined as $\Delta v / \Delta t$ with unit m s^{-2} .
1	Force defined ($F = ma$ / rate of change of momentum) with unit N or kg m s^{-2} .

1(B)(b)**[2 marks]**

Do force and acceleration always point in the same direction? Explain.

ANSWER

Yes for the net force: a is always in the direction of F_{net} ($F = ma$, $m > 0$). No for individual forces, e.g. friction opposes the acceleration of a pulled box.

WORKING

The net force and the acceleration always point in the same direction.

Newton's second law says the rate of change of momentum is proportional to the net applied force *and is in the same direction*. Since $F = ma$ and the mass m is a positive scalar, a points along F_{net} .

An individual force does not have to point along the acceleration. In the notes' example of a box pulled across a table, friction points backwards while the box accelerates forwards. The acceleration follows the *net* force ($101 \text{ N} - 78.4 \text{ N}$, forwards), not each separate force.

Acceleration also doesn't have to point along the *velocity*. A decelerating object (like the glacier below) has its acceleration and net force opposite to its motion.

Marks	Awarded for
1	States that acceleration is always in the direction of the net force, justified with $F = ma$.
1	Distinguishes individual forces (or velocity) from the net force, with an example.

1(B)(c)**[1 mark]**

Glacier velocity 17 km/year. Assuming constant speed for the 90 days of summer, calculate the total distance travelled, in metres.

ANSWER

$$x \approx 4.2 \times 10^3 \text{ m (4192 m)}$$

WORKING

At constant velocity, $x = v t$. 90 days is 90/365 of a year:

$$x = 17\,000 \text{ m/year} \times (90/365) \text{ year} = 4192 \text{ m}$$

Marks**Awarded for****1**

Correct distance in metres.

1(B)(d)**[2 marks]**

The glacier then decelerates at a constant rate and stops after 30 days. Calculate the deceleration in SI units.

ANSWER

$$a = -2.1 \times 10^{-10} \text{ m s}^{-2} \text{ (deceleration of } 2.1 \times 10^{-10} \text{ m s}^{-2}\text{)}$$

WORKING

First convert the velocity to SI units:

$$v_0 = 17\,000 \text{ m} \div (365 \times 24 \times 3600 \text{ s}) = 5.39 \times 10^{-4} \text{ m s}^{-1}$$

Constant acceleration from v_0 to $v = 0$ over $t = 30 \text{ days} = 30 \times 86\,400 \text{ s} = 2.592 \times 10^6 \text{ s}$. Using $v = v_0 + at$:

$$a = (0 - 5.39 \times 10^{-4}) \div 2.592 \times 10^6 = -2.08 \times 10^{-10} \text{ m s}^{-2}$$

Marks	Awarded for
1	Velocity and time both converted to SI (m s^{-1} , s).
1	Correct deceleration with unit m s^{-2} .

1(B)(e)**[3 marks]**

The deceleration was caused by an increase in the coefficient of friction between the glacier and the ground. Calculate the coefficient of friction required to provide this deceleration.

ANSWER

$$\mu_k \approx 2.1 \times 10^{-11} \text{ (unitless)}$$

WORKING

The glacier is moving, so kinetic friction applies: $F_{\text{friction}} = \mu_k N$. On flat ground the normal force equals the weight, $N = mg$.

Friction is the net force causing the deceleration, so by Newton's second law:

$$\mu_k m g = m |a| \Rightarrow \mu_k = |a| / g$$

$$\mu_k = 2.08 \times 10^{-10} \div 9.81 = 2.1 \times 10^{-11}$$

The mass cancels, so you don't need the glacier's mass. μ has no unit.

Marks	Awarded for
1	Friction written as μN with $N = mg$.
1	Friction set equal to ma (Newton's 2nd law) so that m cancels.
1	Correct $\mu \approx 2.1 \times 10^{-11}$, no unit (own-figure from (d) accepted).

Question 1(B) Total: 10 Marks

Question 2 · Option A

Uniform circular motion, the centrifuge, kinetic energy — 10 marks

2(A)(a)

[2 marks]

An object in uniform circular motion has a constantly changing velocity. By referencing Newton's second law, explain why such an object must be experiencing a force.

ANSWER

Changing direction $\Rightarrow$ changing velocity $\Rightarrow$ acceleration $\Rightarrow$ by $F = ma$ there must be a net (centripetal) force.

WORKING

Velocity is a vector. In uniform circular motion the speed is constant but the **direction** of the velocity changes all the time, so the velocity is changing. A changing velocity means the object is **accelerating** ($a = \Delta v / \Delta t \neq 0$).

Newton's second law, $F_{\text{net}} = ma$, says an acceleration can only happen when a **net force** acts on the mass. So the object must be experiencing a force, directed towards the centre of the circle: the centripetal force. (By Newton's first law, with no force it would carry on in a straight line.)

Marks	Awarded for
1	Velocity changes because its direction changes, so the object accelerates.
1	Links acceleration to a net force through $F = ma$.

2(A)(b)**[1 mark]**

The centrifuge spins at 55 rotations per second. How long does it take to complete 1 full rotation?

ANSWER

$T = 0.018 \text{ s (18.2 ms)}$

WORKING

55 rotations per second is the frequency $f = 55 \text{ s}^{-1}$. The period is $T = 1/f$:

$$T = 1 / 55 \text{ s}^{-1} = 0.0182 \text{ s}$$

Marks**Awarded for****1** $T = 1/f = 0.018 \text{ s}.$

2(A)(c)**[2 marks]**

What is the acceleration felt by a sample 8 cm from the axis of rotation?

ANSWER

$a \approx 9.6 \times 10^3 \text{ m s}^{-2}$ ($\approx 970 \text{ g}$), directed towards the centre

WORKING

Angular velocity: $\omega = 2\pi f = 2\pi \times 55 = 345.6 \text{ rad s}^{-1}$.

The question gives $F_c = m\omega^2 r$. By Newton's second law $a = F/m$, so:

$$a = \omega^2 r = (345.6 \text{ rad s}^{-1})^2 \times 0.08 \text{ m} = 9.55 \times 10^3 \text{ m s}^{-2}$$

That is about 970 g ($9554 \div 9.81$), pointing towards the axis of rotation.

EXAM TECHNIQUE

The most common mistake is using $\omega = 55$ directly. 55 is the frequency in rotations per second. The formula needs ω in rad s^{-1} , so multiply by 2π first.

Marks	Awarded for
1	Correct $\omega = 2\pi f$ (in rad s^{-1}) and $a = \omega^2 r$.
1	Correct value with unit.

2(A)(d)**[3 marks]**

If the tube holding the sample were to break instantly, how far would the sample fly in 1 second? Ignore gravity.

ANSWER

The sample flies ≈ 28 m (27.6 m) in a straight line along the tangent

WORKING

At the moment the tube breaks, the sample's speed is its speed around the circle:

$$v = \omega r = 345.6 \text{ rad s}^{-1} \times 0.08 \text{ m} = 27.6 \text{ m s}^{-1}$$

Once it's released there is no centripetal force (and gravity is ignored), so there is no net force. By Newton's first law it keeps moving in a straight line at constant velocity, along the **tangent** to the circle at the point of release.

$$x = v t = 27.6 \text{ m s}^{-1} \times 1 \text{ s} = 27.6 \text{ m}$$

Marks	Awarded for
1	Tangential speed $v = \omega r$.
1	No net force, so straight-line motion at constant speed (Newton's 1st law).
1	Correct distance ≈ 27.6 m (own-figure from (c) accepted).

2(A)(e)**[2 marks]**

If the sample has a mass of 5 g, calculate its kinetic energy in Joules.

ANSWER

$$E_{\text{kin}} \approx 1.9 \text{ J}$$

WORKING

Convert the mass: $m = 5 \text{ g} = 0.005 \text{ kg}$. Then:

$$E_{\text{kin}} = \frac{1}{2} m v^2 = \frac{1}{2} \times 0.005 \text{ kg} \times (27.65 \text{ m s}^{-1})^2 = 1.91 \text{ J}$$

Marks	Awarded for
1	Mass in kg and $E_{\text{kin}} = \frac{1}{2}mv^2$ with the speed from (d).
1	Correct value in J.

Question 2(A) Total: 10 Marks

Question 2 · Option B

Conservation of energy, work done by air resistance — 10 marks

Given: A pine cone of mass 0.14 kg falls 16 m and lands at 13 m/s.

2(B)(a)

[1 mark]

State the law of conservation of energy and the precondition for its applicability.

ANSWER

$E_{\text{pot}} + E_{\text{kin}} = \text{constant}$, provided only conservative forces act

WORKING

The sum of the kinetic and potential energies (the total mechanical energy) stays constant: $E_{\text{total}} = E_{\text{pot}} + E_{\text{kin}} = \text{constant}$.

Precondition: only **conservative forces** (e.g. gravity, spring force) act. If dissipative forces such as friction act, $E_{\text{initial}} - E_{\text{final}} = W_{\text{friction}}$.

Marks

Awarded for

1

Statement **and** the conservative-forces-only condition.

2(B)(b)**[2 marks]**

Assuming no air friction, what velocity would the pine cone have immediately before touching the ground?

ANSWER

$$v \approx 17.7 \text{ m s}^{-1}$$

WORKING

With no friction, energy is conserved, so the loss in potential energy equals the gain in kinetic energy:

$$m g h = \frac{1}{2} m v^2 \Rightarrow v = \sqrt{2 g h}$$

$$v = \sqrt{2 \times 9.81 \times 16} = \sqrt{313.9} = 17.7 \text{ m s}^{-1}$$

Marks	Awarded for
1	Energy conservation (or free-fall equation) set up correctly.
1	Correct $v \approx 17.7 \text{ m s}^{-1}$.

2(B)(c)**[1 mark]**

Does air resistance do positive, negative, or no work on the pine cone? Briefly explain.

ANSWER

Negative work: the force opposes the displacement

WORKING

Air resistance acts **upwards**, opposite to the downward displacement of the falling cone. By the sign convention for work, a force opposite to the displacement does **negative** work. This fits the result that the cone lands slower (13 m s^{-1}) than the frictionless 17.7 m s^{-1} : air resistance removed energy.

Marks**Awarded for****1**

"Negative", with the reason (force opposite to displacement / energy removed).

2(B)(d)**[3 marks]**

How much work is done on the pine cone by air resistance?

ANSWER **$W_{\text{air}} \approx -10.1 \text{ J}$ (10.1 J of mechanical energy dissipated)****WORKING**Energy at the top (at rest): $E_{\text{initial}} = mgh = 0.14 \times 9.81 \times 16 = 21.97 \text{ J}$.Energy just before landing ($h = 0$): $E_{\text{final}} = \frac{1}{2}mv^2 = \frac{1}{2} \times 0.14 \times 13^2 = 11.83 \text{ J}$.

With a dissipative force, the energy difference is the work done by that force:

$$W_{\text{air}} = E_{\text{final}} - E_{\text{initial}} = 11.83 \text{ J} - 21.97 \text{ J} = -10.1 \text{ J}$$

Marks	Awarded for
1	Initial potential energy $mgh = 22.0 \text{ J}$.
1	Final kinetic energy $\frac{1}{2}mv^2 = 11.8 \text{ J}$.
1	Work = difference, -10.1 J , with the negative sign (or magnitude 10.1 J clearly stated as energy removed).

2(B)(e)**[3 marks]**

What is the average force of air resistance exerted on the pine cone during the fall?

ANSWER **$F_{\text{avg}} \approx 0.63 \text{ N}$, upwards****WORKING**

Work = force × displacement, so the average force over the 16 m fall is:

$$F_{\text{avg}} = |W_{\text{air}}| / h = 10.14 \text{ J} \div 16 \text{ m} = 0.63 \text{ N}$$

It is directed **upwards**, opposite to the motion.

Marks	Awarded for
1	Uses $W = F \times d$ with the work from (d).
1	Correct magnitude $\approx 0.63 \text{ N}$ (own-figure accepted).
1	Direction stated: upwards / opposite to the motion.

Question 2(B) Total: 10 Marks

Question 3 · Option A

Specific heat capacity, radiation, calorimetry — 10 marks

3(A)(a)

[1 mark]

Define the quantity “specific heat capacity” and state its SI units.

ANSWER

c = heat per unit mass per kelvin; SI unit $\text{J kg}^{-1} \text{K}^{-1}$

WORKING

Specific heat capacity, c , is the heat required per unit mass to change the temperature of a substance by one degree (1 K). It comes from $Q = mc\Delta T$, so $c = Q/(m\Delta T)$.

Marks

Awarded for

1

Definition **and** unit $\text{J kg}^{-1} \text{K}^{-1}$.

3(A)(b)**[2 marks]**

The most important heat-loss mechanism for a person in hospital is radiation. What is the fractional change in the power radiated when their skin temperature drops from 37 °C to 36 °C?

ANSWER

Fractional change ≈ -0.013 , i.e. the radiated power falls by about 1.3%

WORKING

Radiated power (Stefan-Boltzmann law): $P = A \epsilon \sigma T^4$. A , ϵ and σ don't change, so $P \propto T^4$, and **T must be in kelvin**:

$$T_1 = 37 + 273.15 = 310.15 \text{ K}; T_2 = 36 + 273.15 = 309.15 \text{ K}$$

$$P_2/P_1 = (309.15/310.15)^4 = 0.9872$$

$$\text{Fractional change} = (P_2 - P_1)/P_1 = 0.9872 - 1 = -0.0128$$

EXAM TECHNIQUE

Using °C in the T^4 ratio, $(36/37)^4 = 0.896$, gives a 10% drop. That is wrong and loses the method mark. The T^4 law only works on the absolute (kelvin) scale.

Marks	Awarded for
1	$P \propto T^4$ with temperatures converted to kelvin.
1	Correct fractional change ≈ -0.013 (1.3% decrease).

3(A)(c)**[2 marks]**

Specific heat capacity of the body is $3.5 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$. What total heat must be removed to drop the whole-body temperature of a 65 kg person by $1.0 \text{ }^\circ\text{C}$?

ANSWER

$Q \approx 2.3 \times 10^5 \text{ J (227.5 kJ)}$

WORKING

A temperature difference of $1 \text{ }^\circ\text{C}$ equals 1 K . Then:

$$Q = m c \Delta T = 65 \text{ kg} \times 3.5 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1} \times 1.0 \text{ K} = 2.28 \times 10^5 \text{ J}$$

Marks	Awarded for
1	$Q = mc\Delta T$ applied.
1	Correct value in J.

3(A)(d)**[1 mark]**

What is the average rate of heat loss, assuming the whole body's temperature changes by the same amount as the core temperature?

ANSWER

$P \approx 95 \text{ W}$

WORKING

The core drops from 37 °C to 34 °C, so $\Delta T = 3 \text{ K}$. The total heat removed over the 2 hours is:

$$Q = 65 \times 3.5 \times 10^3 \times 3 = 6.83 \times 10^5 \text{ J}$$

Rate of heat loss = power = energy $\div$ time, with $t = 2 \text{ h} = 7200 \text{ s}$:

$$P = 6.83 \times 10^5 \text{ J} \div 7200 \text{ s} = 94.8 \text{ W}$$

Marks**Awarded for****1**Uses the full 3 K drop over 7200 s, giving $\approx 95 \text{ W}$.

3(A)(e)**[4 marks]**

The person is placed in a thermally isolated bath of 200 litres of water. If the water and body temperatures are the same at the end, what was the initial temperature of the water? ($c_{\text{water}} = 4200 \text{ J kg}^{-1} \text{ K}^{-1}$)

ANSWER

Initial water temperature $\approx 33.2 \text{ }^{\circ}\text{C}$

WORKING

The bath is thermally isolated, so the heat lost by the body equals the heat gained by the water: $Q_{\text{gained}} = -Q_{\text{lost}}$.

Heat lost by the body ($37 \rightarrow 34 \text{ }^{\circ}\text{C}$) = $6.83 \times 10^5 \text{ J}$, from (d). Mass of water: $200 \text{ L} = 0.200 \text{ m}^3$, so $m_{\text{w}} = 200 \text{ kg}$.

$$\Delta T_{\text{water}} = 6.825 \times 10^5 \div (200 \times 4200) = 0.81 \text{ K}$$

Both end at the body's final temperature, $34 \text{ }^{\circ}\text{C}$, so:

$$T_{\text{w,initial}} = 34 \text{ }^{\circ}\text{C} - 0.81 \text{ }^{\circ}\text{C} = 33.2 \text{ }^{\circ}\text{C}$$

EXAM TECHNIQUE

Sanity-check the direction. The water gains heat, so it must start *colder* than $34 \text{ }^{\circ}\text{C}$. An answer above $34 \text{ }^{\circ}\text{C}$ means a sign error.

Marks	Awarded for
1	Heat balance: heat lost by body = heat gained by water.
1	Correct body heat ($6.8 \times 10^5 \text{ J}$) and water mass (200 kg).
1	Correct temperature rise of the water ($\approx 0.81 \text{ K}$).
1	Correct initial temperature $\approx 33.2 \text{ }^{\circ}\text{C}$.

Question 3(A) Total: 10 Marks

Question 3 · Option B

Pascal's principle, hydrostatic pressure, the saline drip, hydraulic lift — 10 marks

Given: Gauge pressure in a vein is $2.4 \times 10^3 \text{ Pa}$; saline density $\rho = 1 \text{ g cm}^{-3}$; bag surface area 0.04 m^2 .

3(B)(a)

[2 marks]

State Pascal's Principle.

ANSWER

Pressure applied to an enclosed fluid is transmitted undiminished to every part of the fluid and to the container walls.

WORKING

The pressure applied at one point in an enclosed fluid is transmitted **undiminished** to every part of the fluid and to the walls of the container.

Marks	Awarded for
1	Pressure applied to an enclosed fluid is transmitted to every part of it.
1	“Undiminished” (equal everywhere), including to the container walls.

3(B)(b)**[2 marks]**

Calculate the minimum height the drip must be above the insertion point for the saline solution to enter the vein.

ANSWER

$$h_{\min} \approx 0.24 \text{ m (24 cm)}$$

WORKING

The saline can only flow in if its hydrostatic pressure at least equals the gauge pressure in the vein: $\rho gh \geq P_{\text{vein}}$. First convert the density: $\rho = 1 \text{ g cm}^{-3} = 1000 \text{ kg m}^{-3}$.

$$h = P / (\rho g) = 2400 \text{ Pa} \div (1000 \text{ kg m}^{-3} \times 9.81 \text{ m s}^{-2}) = 0.245 \text{ m}$$

Marks	Awarded for
1	$\rho gh = P_{\text{vein}}$ with ρ in kg m^{-3} .
1	Correct height $\approx 0.24 \text{ m}$.

3(B)(c)**[2 marks]**

Instead, the drip is suspended 0.2 m above the insertion point. What is the total pressure exerted by the saline at the insertion point?

ANSWER

$$P \approx 1.96 \times 10^3 \text{ Pa (gauge)}$$

WORKING

Hydrostatic pressure from a 0.2 m column of saline:

$$P = \rho g h = 1000 \times 9.81 \times 0.2 = 1.96 \times 10^3 \text{ Pa}$$

This is a gauge pressure, like the $2.4 \times 10^3 \text{ Pa}$ vein value it will be compared against. It is less than the vein pressure, so the saline will not flow in by itself.

EXAM TECHNIQUE

Because the vein pressure is quoted as a gauge pressure, give the saline pressure as a gauge pressure too, so you can compare them directly in (d). If you want to add atmospheric pressure, state it separately: $P_{\text{abs}} = 1.013 \times 10^5 + 1962 \approx 1.03 \times 10^5 \text{ Pa}$ ($P_{\text{B}} = P_{\text{atm}} + \rho g h$).

Marks	Awarded for
1	$P = \rho g h$ with $h = 0.2 \text{ m}$.
1	Correct value $\approx 1960 \text{ Pa}$.

3(B)(d)**[2 marks]**

The additional pressure could be provided by squeezing the bag. What force must be applied to produce this pressure? (Force distributed over the whole 0.04 m^2 bag.)

ANSWER

$F \approx 17.5 \text{ N}$

WORKING

Additional pressure needed: $\Delta P = 2400 - 1962 = 438 \text{ Pa}$.

Pressure is force per area, so $F = P A$:

$$F = 438 \text{ Pa} \times 0.04 \text{ m}^2 = 17.5 \text{ N}$$

Marks	Awarded for
1	Additional pressure $\Delta P = 438 \text{ Pa}$.
1	$F = \Delta P \times A \approx 17.5 \text{ N}$ (own-figure from (c) accepted).

3(B)(e)**[2 marks]**

Alternatively, the bed could be lowered with a hydraulic lift: bed piston area 0.12 m^2 , foot piston area 0.03 m^2 . How much does the foot piston's height increase to drop the bed by the required height?

ANSWER

The foot piston rises $\approx 0.18 \text{ m}$ (18 cm)

WORKING

Lowering the bed increases the height of the drip above the insertion point. It has to go from 0.20 m to the 0.245 m found in (b), so the bed must drop by:

$$\Delta h_{\text{bed}} = 0.2446 - 0.20 = 0.0446 \text{ m}$$

The fluid is incompressible, so the volume leaving the bed cylinder equals the volume entering the foot cylinder:

$$A_{\text{bed}} \Delta h_{\text{bed}} = A_{\text{foot}} \Delta h_{\text{foot}} \Rightarrow \Delta h_{\text{foot}} = (0.12/0.03) \times 0.0446 \text{ m} = 0.18 \text{ m}$$

Marks	Awarded for
1	Required drop of the bed $\approx 0.045 \text{ m}$, from (b) – 0.2 m .
1	Equal volumes, giving $\Delta h_{\text{foot}} \approx 0.18 \text{ m}$ (own-figure accepted).

Question 3(B) Total: 10 Marks

Question 4 · Option A

Electric field, charge density and energy of an ion crossing a neuron membrane — 10 marks

Given: Membrane thickness $0.12 \mu\text{m}$; potential difference between outside and inside -70 mV ; the inside and outside walls are oppositely charged.

4(A)(a)

[1 mark]

What is the SI unit for electric field strength?

ANSWER

V m^{-1} (equivalently N C^{-1})

WORKING

The work done per unit charge moving a distance d along E is $W/q = Fd/q = Ed = V$.
So $E = V/d$, and its unit is volts per metre. (Equivalently newtons per coulomb, since $F = qE$.)

Marks

Awarded for

1

V m^{-1} (or N C^{-1}).

4(A)(b)**[2 marks]**

Calculate the electric field inside the cell membrane, in SI units.

ANSWER

$|E| \approx 5.8 \times 10^5 \text{ V m}^{-1}$, directed from outside to inside

WORKING

The membrane behaves like the two plates of a capacitor, with a uniform field between them. So $V = Ed$:

$$E = V / d = 0.070 \text{ V} \div (0.12 \times 10^{-6} \text{ m}) = 5.8 \times 10^5 \text{ V m}^{-1}$$

The field points from the positive outer wall towards the negative inner wall, i.e. **into** the cell.

Marks	Awarded for
1	$E = V/d$ with μm and mV converted to SI.
1	Correct magnitude with unit.

4(A)(c)**[1 mark]**

Calculate the magnitude of the charge density on the inside of the membrane wall, in C/m².

ANSWER

$$\sigma \approx 5.2 \times 10^{-6} \text{ C m}^{-2}$$

WORKING

For two charged plates the notes give $E = Q/(\epsilon_0 A)$, where Q/A is the charge per unit area (σ). So:

$$\sigma = \epsilon_0 E = 8.854 \times 10^{-12} \times 5.83 \times 10^5 = 5.2 \times 10^{-6} \text{ C m}^{-2}$$

Marks**Awarded for****1** $\sigma = \epsilon_0 E \approx 5.2 \times 10^{-6} \text{ C m}^{-2}$ (own-figure from (b) accepted).

4(A)(d)**[3 marks]**

A positive sodium ion ($m = 3.81 \times 10^{-26} \text{ kg}$, $q = 1.6 \times 10^{-19} \text{ C}$) starts from rest on the outside of the membrane wall. Calculate the change in its potential energy once it reaches the inside wall.

ANSWER

$$\Delta U = -1.1 \times 10^{-20} \text{ J (a decrease of } 1.12 \times 10^{-20} \text{ J)}$$

WORKING

The work done moving a charge through a potential difference is $W = Vq$, so the change in electric potential energy is $\Delta U = q\Delta V$.

The outside wall is positive and the inside wall negative, so the inside is at a potential 70 mV **lower**. Going from outside to inside, $\Delta V = -0.070 \text{ V}$:

$$\Delta U = q \Delta V = 1.6 \times 10^{-19} \text{ C} \times (-0.070 \text{ V}) = -1.12 \times 10^{-20} \text{ J}$$

The ion moves, as the notes put it, “from a region of higher electric PE to lower”, just as a falling object loses gravitational PE.

Marks	Awarded for
1	$\Delta U = q\Delta V$ (or $W = Vq$).
1	Correct magnitude $1.12 \times 10^{-20} \text{ J}$.
1	Correct sign: potential energy decreases (with reason: positive ion moving to lower potential).

4(A)(e)**[3 marks]**

What is the velocity of the sodium ion just before it impacts the inside wall, if it moves through the membrane without encountering any other force?

ANSWER

$v \approx 7.7 \times 10^2 \text{ m s}^{-1}$ (767 m/s), directed into the cell

WORKING

No other force acts, so energy is conserved. The loss in electric PE becomes kinetic energy (the ion starts from rest):

$$\frac{1}{2} m v^2 = |\Delta U| \Rightarrow v = \sqrt{2|\Delta U| / m}$$

$$v = \sqrt{2 \times 1.12 \times 10^{-20} \div 3.81 \times 10^{-26}} = \sqrt{5.88 \times 10^5} = 767 \text{ m s}^{-1}$$

Marks	Awarded for
1	Energy conservation: loss in PE = gain in KE.
1	Rearranged correctly for v.
1	Correct $v \approx 767 \text{ m s}^{-1}$ (own-figure from (d) accepted).

Question 4(A) Total: 10 Marks

Question 4 · Option B

Ohm's law, a two-resistor circuit, electrical power, current as charge flow — 10 marks

Given: 12 V supply; $R_1 = 5 \, \Omega$; current at point a is 2 A; R_1 and R_2 are in a single loop (series), so the same 2 A flows through both.

4(B)(a)

[2 marks]

State and explain Ohm's Law. Give an example of when Ohm's law is applicable.

ANSWER

**V = IR: current is proportional to voltage for a fixed resistance.
Applies e.g. to metal resistors / heating elements.**

WORKING

Ohm's law: $V = IR$. The current I through a conductor is proportional to the potential difference V across it, with the resistance R as the constant of proportionality. Equivalently, for a given V , $I \propto 1/R$.

Explanation: R measures how strongly a material opposes the flow of charge. It depends on the material's resistivity and the conductor's shape, $R = \rho l/A$. Doubling V doubles I as long as R stays constant.

Example: a metal resistor or heating element, e.g. the notes' 2 kW kettle on a 220 V supply, $R = V^2/P = 24 \, \Omega$. Another is the current through human skin in an electric shock, $I = V/R (\approx 0.4 \text{ A through wet skin, } R \approx 500 \, \Omega, \text{ at } 220 \text{ V})$.

Note on the lecture material: The lecture notes use $V = IR$ throughout but don't give a formal written statement of "Ohm's law" or its conditions (e.g. constant temperature). The answer above is built only from how the lecturer uses $V = IR$ and $I \propto 1/R$.

Marks	Awarded for
1	$V = IR$ stated, with $I \propto V$ at constant R explained.
1	Valid example of an ohmic situation (resistor, heating element, skin resistance...).

4(B)(b)**[2 marks]**Calculate the resistance of the second resistor, R_2 .**ANSWER**

$R_2 = 1 \, \Omega$

WORKING

The two resistors form one loop, so the same current (2 A) flows through both.

Voltage across R_1 : $V_1 = I R_1 = 2 \, \text{A} \times 5 \, \Omega = 10 \, \text{V}$.

V is the work done per unit charge moving it through the circuit, and the battery provides 12 J per coulomb. So the rest must be used across R_2 : $V_2 = 12 - 10 = 2 \, \text{V}$.

$$R_2 = V_2 / I = 2 \, \text{V} \div 2 \, \text{A} = 1 \, \Omega$$

(Check: total resistance = $12 \, \text{V} \div 2 \, \text{A} = 6 \, \Omega = 5 \, \Omega + 1 \, \Omega$.)

Marks	Awarded for
1	Correct method: $V_1 = I R_1$ and the remaining voltage across R_2 (or $R_{\text{total}} = V/I = 6 \, \Omega$).
1	$R_2 = 1 \, \Omega$.

4(B)(c)**[2 marks]**

Calculate the magnitude of the voltage across each of the resistors.

ANSWER

$$V_1 = 10 \text{ V}; V_2 = 2 \text{ V}$$

WORKING

$$V_1 = I R_1 = 2 \text{ A} \times 5 \Omega = 10 \text{ V}$$

$$V_2 = I R_2 = 2 \text{ A} \times 1 \Omega = 2 \text{ V}$$

Check: $10 \text{ V} + 2 \text{ V} = 12 \text{ V}$, the supply voltage.

Marks	Awarded for
1	$V_1 = 10 \text{ V}$.
1	$V_2 = 2 \text{ V}$ (own-figure from (b) accepted).

4(B)(d)**[2 marks]**

Calculate the power used by each resistor.

ANSWER

$P_1 = 20 \text{ W}; P_2 = 4 \text{ W}$

WORKING

$$P_1 = I^2 R_1 = (2 \text{ A})^2 \times 5 \Omega = 20 \text{ W (or } P = IV = 2 \times 10)$$

$$P_2 = I^2 R_2 = (2 \text{ A})^2 \times 1 \Omega = 4 \text{ W (or } P = IV = 2 \times 2)$$

Check: total $24 \text{ W} = 12 \text{ V} \times 2 \text{ A}$, the power delivered by the supply. In each resistor this energy is dissipated as heat.

Marks	Awarded for
1	$P_1 = 20 \text{ W}.$
1	$P_2 = 4 \text{ W}.$

4(B)(e)**[2 marks]**

Calculate the number of electrons that flow past the point marked a in 1 second.

ANSWER **$N \approx 1.25 \times 10^{19}$ electrons per second****WORKING**

Current is charge per unit time, $I = Q/t$, so in 1 s: $Q = I t = 2 \text{ A} \times 1 \text{ s} = 2 \text{ C}$.

Each electron carries $e = 1.602 \times 10^{-19} \text{ C}$:

$$N = Q / e = 2 \text{ C} \div 1.602 \times 10^{-19} \text{ C} = 1.25 \times 10^{19} \text{ electrons}$$

Marks	Awarded for
1	$Q = It = 2 \text{ C}$.
1	$N = Q/e \approx 1.25 \times 10^{19}$.

Question 4(B) Total: 10 Marks

Question 5 · Option A

Earth's magnetic poles, Lorentz force, circular motion of a charge — 10 marks

Given: $q = +5 \text{ mC}$, $m = 1 \times 10^{-10} \text{ kg}$, $v = 3 \times 10^4 \text{ m s}^{-1}$ left $\rightarrow$ right, $B = 5 \text{ } \mu\text{T}$ into the page.

5(A)(a)

[2 marks]

The magnetic north pole on a compass points towards the geographic north pole. Explain why this means the Earth's magnetic south pole must lie in the direction of the Earth's geographic north pole.

ANSWER

Unlike poles attract. The compass's N pole is pulled north, so the pole there must be a magnetic S pole.

WORKING

A compass is a dipole magnet with a north and a south pole. With magnets, **similar poles repel and unlike poles attract**.

The compass's north pole is attracted towards the geographic north. Only an unlike pole, a **magnetic south pole**, would attract a north pole. So the Earth's magnetic south pole must lie near the geographic north pole, and vice versa. Put another way, a compass needle lines up along the field B , which runs *into* a south pole.

Marks	Awarded for
1	Unlike poles attract (compass is a magnetic dipole).
1	So the pole that attracts the compass's N end must be a magnetic S pole, at geographic north.

5(A)(b)**[2 marks]**

Draw a copy of the diagram and include the direction of the Lorentz force that the particle feels.

ANSWER

Lorentz force points up the page (perpendicular to v , in the plane of the page)

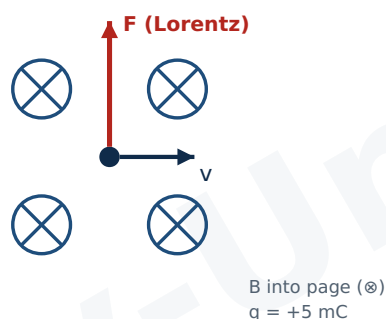
WORKING

Fig. 1 — Positive charge moving right through B into the page: the Lorentz force points up the page (towards the top), perpendicular to both v and B.

The force on a moving charge is perpendicular to **both** v and B. Using the Left Hand Rule: first finger along B (into page), second finger along the current (to the right, since q is positive), and the thumb gives F , pointing **up the page**.

This matches the lecture example where a $+Q$ moving up the page in a field into the page is pulled to the left. Rotating that picture by 90° gives this case.

Marks	Awarded for
1	Diagram reproduced with B into page and v to the right.
1	Force correctly drawn up the page, perpendicular to v .

5(A)(c)**[2 marks]**

Calculate the magnitude of the Lorentz force experienced by the particle.

ANSWER

$F = 7.5 \times 10^{-4} \text{ N}$

WORKING

The velocity is perpendicular to B, so:

$$F = B Q v = 5 \times 10^{-6} \text{ T} \times 5 \times 10^{-3} \text{ C} \times 3 \times 10^4 \text{ m s}^{-1} = 7.5 \times 10^{-4} \text{ N}$$

Marks	Awarded for
1	$F = BQv$ with μT and mC converted.
1	Correct value $7.5 \times 10^{-4} \text{ N}$.

5(A)(d)**[3 marks]**

The particle undergoes uniform circular motion in the field. Calculate the radius of this motion.

ANSWER

$$r = 120 \text{ m}$$

WORKING

The Lorentz force provides the centripetal force:

$$m v^2 / r = B Q v \Rightarrow r = m v / (Q B)$$

$$r = (1 \times 10^{-10} \times 3 \times 10^4) \div (5 \times 10^{-3} \times 5 \times 10^{-6}) = 3 \times 10^{-6} \div 2.5 \times 10^{-8} = 120 \text{ m}$$

Marks	Awarded for
1	Lorentz force set equal to centripetal force.
1	Rearranged to $r = mv/(QB)$.
1	$r = 120 \text{ m}$.

5(A)(e)**[1 mark]**

If the particle were replaced with one of opposite charge moving in the opposite direction, would the direction of the Lorentz force stay the same or change? Explain briefly.

ANSWER

It stays the same: the two reversals cancel

WORKING

Reversing the charge's sign reverses the force. Reversing the velocity also reverses the force. Both happen, so the two reversals cancel. Equivalently, a negative charge moving left is the same current as a positive charge moving right.

Marks**Awarded for****1**

"Same", with the double-reversal / equivalent-current reasoning.

Question 5(A) Total: 10 Marks

Question 5 · Option B

Magnetic field of a wire, resistivity, force between parallel currents — 10 marks

Given: Two parallel 5 m wires, 2 m apart, each on a 12 V supply, each carrying 2.5 A in the same direction; wire radius 2 mm.

5(B)(a)

[2 marks]

Draw a diagram of this setup, clearly labelling the direction of the current in each wire and the direction of the magnetic field on both sides of each wire.

ANSWER

Currents drawn parallel and in the same direction; B out of page on the left and into page on the right of each wire

WORKING

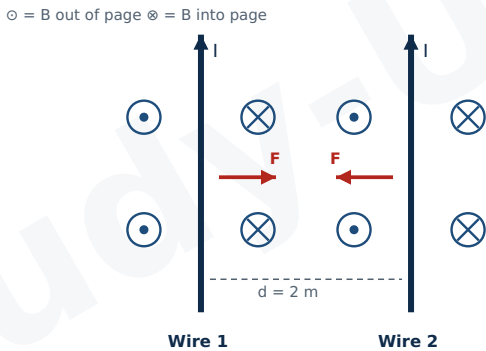


Fig. 2 — Both currents flow up the page. Right-hand rule: B comes out of the page (⊙) on the left of each wire and goes into the page (⊗) on the right. The force between the wires is attractive.

Right-hand rule: grip the wire with your right hand, thumb along I (up the page). Your fingers curl in the direction of B , which circles the wire: **out of the page on the left** and **into the page on the right** of each wire.

Marks	Awarded for
1	Both currents shown in the same direction.
1	B directions correct on both sides of both wires.

5(B)(b)**[3 marks]**

Calculate the resistivity of the material the wires are made of.

ANSWER

$$\rho \approx 1.2 \times 10^{-5} \Omega \text{ m}$$

WORKING

Resistance of each wire: $R = V/I = 12 \text{ V} \div 2.5 \text{ A} = 4.8 \Omega$.

Cross-sectional area: $A = \pi r^2 = \pi \times (2 \times 10^{-3} \text{ m})^2 = 1.26 \times 10^{-5} \text{ m}^2$.

From $R = \rho \ell / A$:

$$\rho = R A / \ell = 4.8 \Omega \times 1.26 \times 10^{-5} \text{ m}^2 \div 5 \text{ m} = 1.2 \times 10^{-5} \Omega \text{ m}$$

Marks	Awarded for
1	$R = V/I = 4.8 \Omega$.
1	$A = \pi r^2$ in m^2 .
1	$\rho = RA/\ell \approx 1.2 \times 10^{-5} \Omega \text{ m}$.

5(B)(c)**[2 marks]**

Calculate the magnitude of the magnetic field strength, B , at the location of wire 2 due to the current in wire 1.

ANSWER

$$B = 2.5 \times 10^{-7} \text{ T}$$

WORKING

$$B = (\mu_0 / 2\pi) \cdot I / d = (4\pi \times 10^{-7} / 2\pi) \times 2.5 \text{ A} \div 2 \text{ m} = 2 \times 10^{-7} \times 1.25$$

$$B = 2.5 \times 10^{-7} \text{ T}$$

Marks	Awarded for
1	Correct formula $B = \mu_0 I / (2\pi d)$.
1	$B = 2.5 \times 10^{-7} \text{ T}$.

5(B)(d)**[2 marks]**

Calculate the magnitude of the force exerted on each wire.

ANSWER

$F \approx 3.1 \times 10^{-6} \text{ N}$ on each wire

WORKING

Wire 2 carries I_2 perpendicular to the field B from wire 1, over the full length $\ell = 5 \text{ m}$:

$$F = B I \ell = 2.5 \times 10^{-7} \text{ T} \times 2.5 \text{ A} \times 5 \text{ m} = 3.1 \times 10^{-6} \text{ N}$$

Equivalently $F/\ell = (\mu_0/2\pi) \cdot I^2/d = 6.25 \times 10^{-7} \text{ N m}^{-1}$, $\times 5 \text{ m}$. By Newton's third law wire 1 feels the same magnitude.

Marks	Awarded for
1	$F = BI\ell$ (or F/ℓ formula) with $\ell = 5 \text{ m}$.
1	$F \approx 3.1 \times 10^{-6} \text{ N}$ (own-figure from (c) accepted).

5(B)(e)**[1 mark]**

Do the wires move towards each other or away from each other due to this force?

ANSWER

Towards each other (attractive)

WORKING

For currents flowing in the **same** direction, the Lorentz force on each wire points towards the other wire, so the force is attractive. (Opposite currents would repel.)

Marks	Awarded for
1	Towards each other.

Question 5(B) Total: 10 Marks

Question 6 · Option A

Transverse vs longitudinal waves, $v = f\lambda$, two-source interference — 10 marks

Given: Speakers $D = 2.01\text{ m}$ apart, in phase, $\lambda = 0.9\text{ m}$; microphone $d_1 = 1.8\text{ m}$ directly in front of speaker 1.

6(A)(a)

[2 marks]

What is the difference between transverse waves and longitudinal waves? Give an example of each.

ANSWER

Transverse: oscillation $\perp$ to propagation (e.g. light, water waves). Longitudinal: oscillation $\parallel$ to propagation (e.g. sound).

WORKING

Transverse wave: the medium moves **perpendicular** to the direction the wave travels. Examples: water waves, light (and all electromagnetic waves).

Longitudinal wave: the medium vibrates **parallel** to the direction the wave travels, creating regions of compression and rarefaction. Example: sound waves in air.

Marks	Awarded for
1	Correct distinction ($\perp$ vs $\parallel$ to the direction of propagation).
1	A correct example of each.

6(A)(b)**[1 mark]**

What is the frequency of the sound wave? (Speed of sound = 343 m s^{-1} .)

ANSWER **$f \approx 381 \text{ Hz}$** **WORKING**

$$v = f \lambda \Rightarrow f = v / \lambda = 343 \text{ m s}^{-1} \div 0.9 \text{ m} = 381 \text{ Hz}$$

Marks**Awarded for****1** $f \approx 381 \text{ Hz}.$

6(A)(c)**[2 marks]**Calculate the distance d_2 , in metres.**ANSWER**

$d_2 \approx 2.70 \text{ m (2.698 m)}$

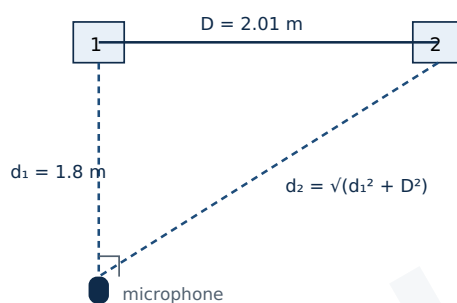
WORKING

Fig. 3 — d_1 and D are at right angles, so d_2 is the hypotenuse.

The microphone is directly in front of speaker 1, so $d_1 \perp D$ and d_2 is the hypotenuse:

$$d_2 = \sqrt{d_1^2 + D^2} = \sqrt{(1.8^2 + 2.01^2)} = \sqrt{(3.24 + 4.04)} = \sqrt{7.28} = 2.70 \text{ m}$$

Marks	Awarded for
1	Pythagoras set up correctly.
1	$d_2 \approx 2.70 \text{ m}$.

6(A)(d)**[2 marks]**

What is the condition for constructive and destructive interference at the microphone in relation to the difference $d_2 - d_1$?

ANSWER

Constructive: $d_2 - d_1 = n\lambda$. **Destructive:** $d_2 - d_1 = (n + \frac{1}{2})\lambda$.

WORKING

The speakers emit in phase, so what matters is whether the two waves **arrive** in phase.

Constructive: the waves arrive in phase (in step) when the extra distance is a whole number of wavelengths: $d_2 - d_1 = n\lambda$ ($n = 0, 1, 2, \dots$). Crest meets crest and the amplitudes add, up to $2A$.

Destructive: the waves arrive exactly out of phase when the extra distance is an odd number of half-wavelengths: $d_2 - d_1 = (n + \frac{1}{2})\lambda$. Crest meets trough and the waves cancel.

Note on the lecture material: The notes show that in-phase waves add constructively and exactly out-of-phase waves cancel, but they don't write out the path-difference conditions $n\lambda$ and $(n + \frac{1}{2})\lambda$. The conditions above follow directly from that, together with the lecture definition of λ as the crest-to-crest distance.

Marks	Awarded for
1	Constructive: path difference = whole number of wavelengths.
1	Destructive: path difference = odd number of half-wavelengths.

6(A)(e)**[2 marks]**

Calculate whether the sound waves interfere constructively or destructively at the microphone.

ANSWER

Path difference $\approx 1\lambda \Rightarrow$ constructive interference

WORKING

$$d_2 - d_1 = 2.698 - 1.8 = 0.898 \text{ m}$$

$$(d_2 - d_1) / \lambda = 0.898 \div 0.9 = 1.00$$

The path difference is (to within rounding) exactly one wavelength, $n = 1$. The waves arrive in phase.

Marks	Awarded for
1	Path difference $\approx 0.90 \text{ m} \approx 1.00\lambda$.
1	Concludes constructive (own-figure from (c) with consistent reasoning accepted).

6(A)(f)**[1 mark]**

Based on your previous answers, will the microphone pick up sounds or not?

ANSWER**Yes — a loud (reinforced) sound****WORKING**

Yes. The interference is constructive, so the two waves add to give a resultant of larger amplitude (up to $2A$) at the microphone. It picks up a **loud** sound, not a silence.

Marks	Awarded for
-------	-------------

1	Yes, loud/reinforced sound, consistent with (e).
---	--

Question 6(A) Total: 10 Marks

Question 6 · Option B

Snell's law, critical angle, refraction through an aquarium wall — 10 marks

6(B)(a)

[3 marks]

State and explain Snell's law of refraction (including the definition of the refractive index).

ANSWER

$n_1 \sin \theta_1 = n_2 \sin \theta_2$ (angles from the normal, rays and normal coplanar); $n = c / v_{\text{medium}}$

WORKING

When a ray passes from medium 1 into medium 2, it is refracted (bent) at the boundary, and:

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

θ_1 is the angle of incidence and θ_2 the angle of refraction, both measured from the **normal**. The incident ray, the refracted ray and the normal all lie in the same plane.

The refractive index of a medium is $n \equiv (\text{speed of light in vacuum, } c) \div (\text{speed of light in the medium})$. n depends on the optical properties of the medium: vacuum = 1, air ≈ 1 , water ≈ 1.33 , glass ≈ 1.5 .

So a ray entering a medium of higher n bends **towards** the normal, and a ray entering a medium of lower n bends **away** from it.

Marks	Awarded for
1	Snell's law stated with angles defined from the normal.
1	Refractive index defined as $n = c/v$.
1	Explanation: bending towards/away from the normal, or rays and normal in the same plane.

6(B)(b)**[1 mark]**

Explain the concept of the “critical” angle in the context of Snell's law.

ANSWER

θ_{crit} = incidence angle (from higher-n to lower-n medium) giving 90° refraction; above it, total internal reflection

WORKING

When light goes from a medium of higher n to lower n ($n_1 > n_2$), it bends away from the normal. The **critical angle** θ_{crit} is the angle of incidence at which the angle of refraction is exactly 90° , so the refracted ray runs along the boundary. For any angle of incidence greater than θ_{crit} , the ray is reflected back into medium 1. This is **total internal reflection** (the basis of optical fibres).

Marks	Awarded for
1	Defined via 90° refraction angle when going from higher to lower n (TIR above it).

6(B)(c)**[1 mark]**

What is the general formula for the critical angle?

ANSWER

$\theta_{\text{crit}} = \sin^{-1}(n_2/n_1)$

WORKING

Put $\theta_2 = 90^\circ$ into Snell's law: $n_1 \sin \theta_{\text{crit}} = n_2 \sin 90^\circ = n_2$, so:

$$\theta_{\text{crit}} = \sin^{-1}(n_2 / n_1), \text{ with } n_1 > n_2$$

Marks	Awarded for
1	$\theta_{\text{crit}} = \sin^{-1}(n_2/n_1)$.

6(B)(d)**[2 marks]**

What is the critical angle for the transition from glass ($n = 1.5$) to air ($n = 1$)?

ANSWER

$$\theta_{\text{crit}} \approx 41.8^\circ$$

WORKING

$$\theta_{\text{crit}} = \sin^{-1}(1 / 1.5) = \sin^{-1}(0.667) = 41.8^\circ$$

Marks	Awarded for
1	Correct substitution $n_2/n_1 = 1/1.5$.
1	$\theta_{\text{crit}} \approx 41.8^\circ$.

6(B)(e)**[3 marks]**

A ray strikes the glass from air at $\theta_1 = 45^\circ$ to the normal. Calculate the angle the light makes with the normal when it enters the water (θ_2 in the figure). $n_{\text{water}} = 1.33$.

ANSWER

$\theta_2 \approx 32.1^\circ$ (angle in glass $\approx 28.1^\circ$)

WORKING

Air $\rightarrow$ glass ($n = 1$ to 1.5):

$$\sin \theta_{\text{glass}} = (1 / 1.5) \times \sin 45^\circ = 0.4714 \Rightarrow \theta_{\text{glass}} = 28.1^\circ$$

The two faces of the glass are parallel, so the ray hits the glass-water face at the same 28.1° to the normal.

Glass $\rightarrow$ water ($n = 1.5$ to 1.33):

$$\sin \theta_2 = (1.5 / 1.33) \times \sin 28.1^\circ = (1.5/1.33) \times 0.4714 = 0.5317 \Rightarrow \theta_2 = 32.1^\circ$$

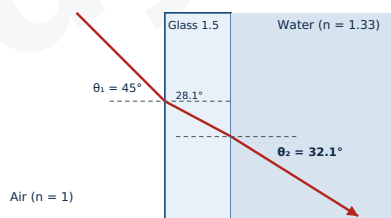


Fig. 4 — Bends towards the normal into glass, away from it into water.

Marks	Awarded for
1	Snell's law at the air-glass boundary: θ in glass $\approx 28.1^\circ$.
1	Uses the same angle at the second (parallel) face and applies Snell's law glass $\rightarrow$ water.
1	$\theta_2 \approx 32.1^\circ$.

Question 6(B) Total: 10 Marks